Extreme Scale Engineering

Architecture Patterns for Scaling Solution to Billion+ Transactions

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About Me

Kishore Vutukuri

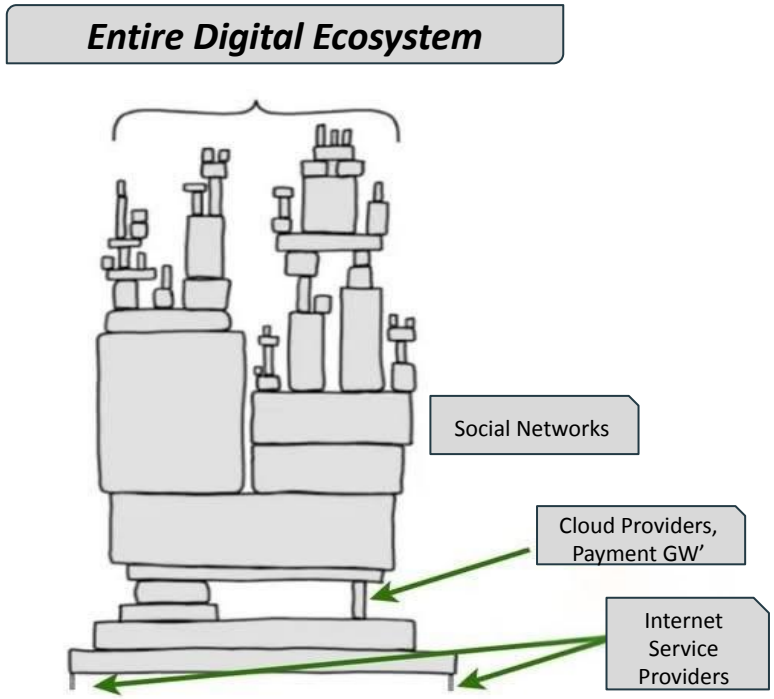
Enterprise Architect @ EY

Tech Enthusiast & Problem Solver

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Backbone of Digital World



Tier 1 Operators in Numbers to Support Nation Scale

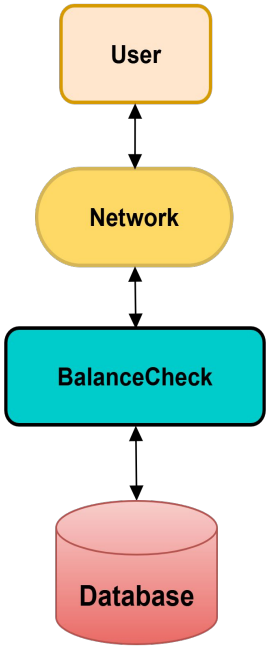
- Tier 1 Operators: ~470 million subscribers and ~375 million subscribers
- India total users: ~1 billion connections
- India consumes 4x more mobile data per user than US.

India is the largest telecom data consumption market in the world.

Meaning: Systems must be engineered for Sale ,failure, concurrency, spikes, and unpredictable traffic patterns.

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Basic Flow & Common Business Needs



Business Needs
Highly available systems
Low Latency and high performant systems
National level population support - Hundreds of millions users and Billion+ daily transactions
Zero Revenue losses.
Continuous and High number of requests.

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Common NFR's

Key Metric	Requirement
Average RPS/TPS	30-50K
Response Times (P99)	Milli/Micro Seconds
Transactions	Billion+ Per Day.
Availability	99.999%
Minimum Load/Minute	Million+ Transactions Throughput

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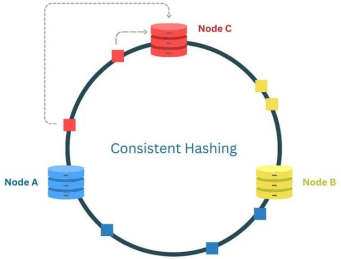
Common Bottlenecks for Scale

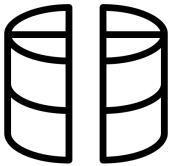
- ✖ SPOF and Single systems become chokepoints
- ✖ System becomes fragile under spikes
- ✖ Latencies are the new downtimes.
- ✖ Synchronous chains create cascading failures
- ✖ DB becomes bottleneck

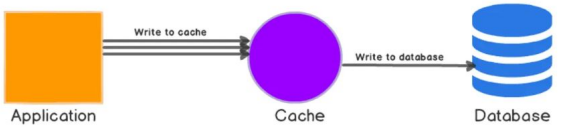
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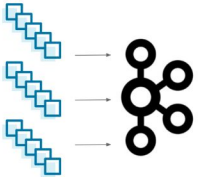
Key Architecture Patterns - Battle Tested

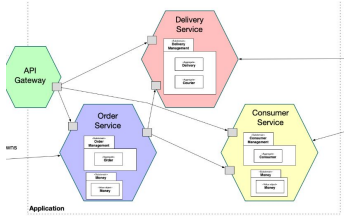
- ✅ Horizontal Scaling
- ✅ Consistent Hashing & Sharding
- ✅ Caching - Write Behind Strategy
- ✅ Event Driven Pattern (Pub/Sub).
- ✅ Microservices & Auto Scaling
- ✅ Active-Active services.





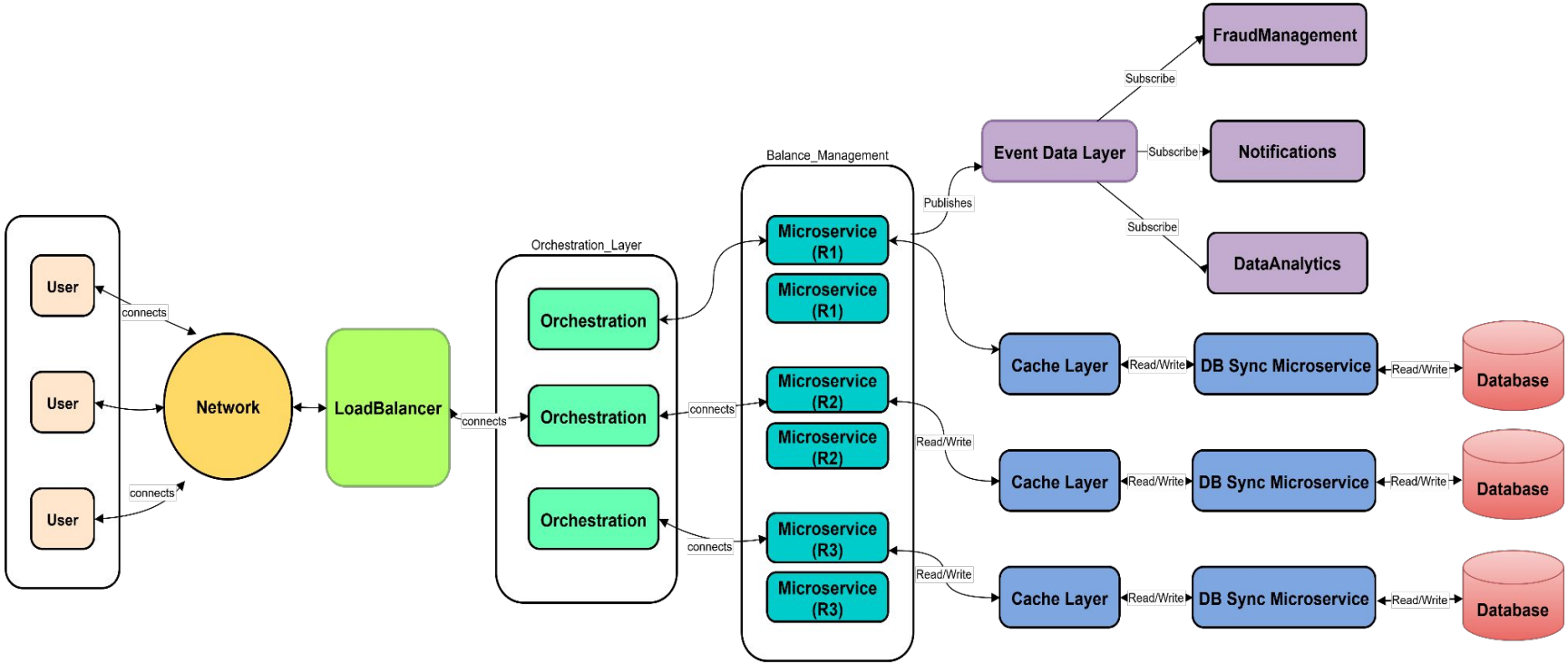






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Extreme Scale - Architecture *



* Illustrative reference

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Additional Important Strategies

Read More Here



- 💪Zero Downtime deployment strategies.
- 💪Resilience patterns
- 💪Disaster Recovery strategy
- 💪Performance and Chaos Testing.
- 💪Archiving & Purging to Secondary Storage

“The key to scaling is to decouple — in architecture and in teams.”

— Werner Vogels (Amazon CTO)*

”If you're afraid to change something it is clearly poorly designed.”

— Martin Fowler(Thoughtworks)*

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